Wave Numbers of Linear Progressive Waves

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Introduction

The local wave numbers for linear progressive waves over constant depth or slowly varying topography can be determined by the wave dispersion relation, which for a wave with period T in water of depth h may be written in dimensionless form as

$$x = y \tanh y \quad (1)$$

in which $x = \sigma^2 h/g$; $\sigma = 2\pi/T$, the radian frequency; g denotes gravity; and $y = kh$, in which k is the wave number. The limiting value of y in deep water is

$$y = x \quad (2)$$

and in shallow water

$$y = \sqrt{x} \quad (3)$$

For wave propagation problems, the positive real root of the dispersion equation is of interest. However, the wave dispersion relation (Eq. 1) is a nonlinear equation. Various methods of approximate solutions, including Eckart's dispersion equation, iterative techniques, and the Pade's asymptotic form, were developed; a review can be found in Dean and Dalrymple (1984). The accuracy and efficiency of the iterative method depend on its initial value. On the other hand, the solution of Pade's form is direct and valid for the entire range of values of kh. Hunt (1979) proposed a polynomial solution by Pade's form, and improved the accuracy by iteratively making perturbations in the coefficients of the series. As a result, Hunt's solution is not unique. Moreover, the Pade's form solution series converges relatively slowly and in general requires a higher degree polynomial.

The current approach is to find the point of inflection of the wave dispersion relation and then to represent the two sections of the dispersion curve by suitable asymptotic series expansions. The solution of wave number in truncated series can be easily obtained on a hand calculator. It is particularly useful for a large number of wave number calculations, such as wave spectral analysis and refraction of ocean waves.

Dispersion Curves and Approximate Solutions

The linear wave dispersion relation can be conveniently written in the form

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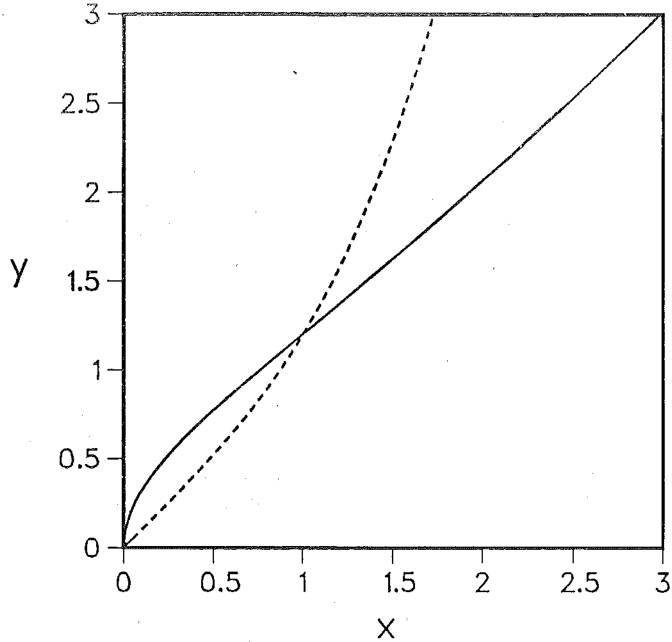

FIG. 1.—Dispersion Curves of Eq. 1 (Solid Curve) and Eq. 4 (Dashed Line)

$$\sqrt{x} = \sqrt{y \tanh y} \quad (4)$$

as used by Mei (1983). The two curves of Eqs. 1 and 4 are shown in Fig. 1. It is seen that Eq. 1 tends to be a straight line when x is greater than 1. When x is close to zero, the curve of Eq. 1 is rather nonlinear and has a slope of infinity. On the other hand, the curve of Eq. 4 has a finite value of slope near $x = 0$, and grows in a quasilinear manner toward $x = 1$. Accordingly, the wave dispersion relation may be favorably represented by an approximate function, which is Eq. 1 when $x > 1$ and Eq. 4 when $x < 1$. The two curves intersect at $x = 1$, where the dispersion curve (Eq. 1) has a point of inflection separating the deep and shallow water regions. By differentiating Eq. 1 twice, the point of inflection is found at $x = 1$ and $y = 1.200$. A similar analysis was attempted by Longuet-Higgins (1972), but the exact value, $x \equiv 1$, was misprinted. It can be shown that at the inflection point the group velocity has a maximum value and the shoaling height is a minimum (see Fig. 2.4 in Wiegel, 1964).

Approximate Solutions.—Two series solutions of kh are derived. In a shallow water region, the tanh function is expanded in a series, noting that Eq. 4 approaches Eq. 3 for $x << 1$. These approximations yield

$$y = \sqrt{x}\left(1 + \frac{x}{6} + Ax^2\right) \quad (5)$$

The higher-order terms in x are absorbed by the coefficient, A, within a

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limited accuracy. Then, by the end condition at $x \equiv 1$ and $y = 1.200$, the approximate solution for Eq. 4 can be written as

$$y = \sqrt{x}\left[1 + \frac{x}{6}\left(1 + \frac{x}{5}\right)\right] \quad \text{(6)}$$

with an accuracy on $y(=kh)$ better than 0.05% for $x < 1$.

In a deep water region, the tanh function approaches 1 and is best described in the exponential form. Using the limiting condition (Eq. 2), we expand Eq. 1 in a series

$$y = x(1 + 2e^{-2x} + 2e^{-4x} + 2e^{-6x} + \ldots) \quad \text{(7)}$$

For practical purposes, a truncated form is empirically proposed

$$y' = x(1 + 1.26e^{-1.84x}) \quad \text{(8)}$$

which satisfies the matching condition ($x = 1$, $y = 1.200$), and reduces to Eq. 2 as x becomes large with an accuracy of 0.2%. To achieve the same accuracy as that given by Eq. 6 and to simplify the solution form, a truncation of Eq. 7 to two terms and substitution of Eq. 8 gives the composite result

$$y = x[1 + 2t(1 + t)] \quad \text{(9)}$$

in which $t = e^{-2y'}$ and y' is given by Eq. 8. It can be shown that Eq. 9 gives an accuracy of 0.01% for $x > 1.0$, and the resulting errors diminish exponentially to zero as x increases.

Range of Validities and Accuracy Comparisons

To establish the applicability of the method, comparisons between the current solution for kh of a progressive wave and other approximate solutions were made for the class of relative depth, h/L_o, in which L_o is the deep water wavelength, ranging from 0 to 1. Results are shown in Table 1. The "exact" solutions were obtained by Newton's iterative method with an absolute error of 10^{-6}, and the deep water value is used as an initial guess. Hunt's solutions were calculated by a six-term Pade's form expansion. The new solutions were calculated by using Eq. 6 for $h/L_o <$

TABLE 1.—Comparisons of Wave Number Calculations

$\frac{h}{L_o}$ (1)	Exact (2)	New (3)	Hunt (4)
0.005	0.17818	0.17818	0.17821
0.01	0.25332	0.25332	0.25338
0.02	0.36210	0.36210	0.36261
0.05	0.59160	0.59168	0.59136
0.1	0.88581	0.88609	0.88571
0.2	1.41446	1.41423	1.41469
0.5	3.15312	3.15312	3.15524
0.8	5.02705	5.02705	5.02836
1.0	6.28332	6.28332	6.28389

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0.2 and Eq. 9 for $h/L_o \geq 0.2$, i.e., approximately $y = \sqrt{2}$ is the criterion for separating deep and shallow water regions. Clearly, the new solutions using two-term series favorably agree with the exact solutions, and give an error of less than 0.05% over the entire ranges of (h/L_o). On the other hand, Hunt's solution using six terms yields no better accuracy than the current results.

Numerical Applications and Efficiency Comparisons

To demonstrate the efficiency of the proposed method, numerical comparisons were made between the current method and Newton's method.

Wave Spectral Analysis.—For nearshore studies, an offshore wave slope array of four pressure sensors is often used to acquire directional wave spectra. In performing the data analysis, the pressure spectra were first constructed from the measured time series response, and then, applying the linear wave theory, the pressure spectra were converted into the spectra of sea surface elevation via a transfer function, $H(k)$, which must be calculated for each spectral component at the depths of the pressure sensors. Numerical calculations were carried out using Newton's iterative method with a tolerance error of 10^{-4}, and the proposed method. The clocked CPU time on the IBM 3033 indicates that the direct calculations can save 30% in comparison to the iterative method.

Wave Refraction.—For ocean wave forecasts, a refraction model is used to transform deep water wave spectra to desired coastal regions. The refraction model requires computations of wave number and wave speed. The local wave speed $c_o = \sigma/k$, relative to its deep water value $c = g/\sigma$, is given by

$$\frac{c}{c_o} = \frac{\sigma^2}{gk} = \frac{x}{y} \quad \text{(10)}$$

Therefore the calculation for wave speed can be easily obtained.

To study the wave transformation in Monterey Bay, the writers employed a finite element refraction model (Wu, et al., 1985) computing shoaling and refraction for a given directional wave spectra. Numerical experiments were performed for 10 frequencies and 5 direction bands. It was found that the total CPU time used by the model with the proposed scheme substituted for the iterative method was reduced by nearly 600 seconds.

Conclusion and Remarks

A direct method of solution for wave numbers of linear progressive waves is presented. The following advantages are summarized:

1. The explicit solution for wave number, as given by Eq. 6 for values of $h/L_o < 0.2$ and Eq. 9 for $h/L_o \geq 0.2$ has an accuracy better than 0.05% over the entire ranges of application. In view of the fact that current hydrographic depth measurements have errors on the order of 0.1% or greater, the current solution should suffice for general wave applications.

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2. The series solution form in Eqs. 6 and 9 is simpler than the Pade's form solution proposed by Hunt (1979), and is apparently more applicable than a tabulated look-up table. Numerical examples demonstrate that for a desired accuracy the current method can save a significant amount of computing time in the wave calculations.

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